Examples for WSM v8.5.0

Vocus Wholesale Web Services

Commercial and in confidence

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1 Basic Examples

This documentation contains examples of various WSM web service requests and responses. The example SOAP XML messages have been simplified to improve readability.

1.1 WSM SOAP Request and Response

There are only 4 operations in WSM: Create, Get, Set, and Delete. All operations use a uniform XML string format. The following example shows the basic XML string construction for WSM:

SOAP Request

```
<WSM::$Request>
    <AccessKey>$AccessKey</AccessKey>
    <ProductID>$ProductID</ProductID>
    <PlanID>$PlanID</PlanID>
    <Scope>$Scope </Scope>
    <Parameters>
        <Param id="$ParamName_01">$ParamContent_01</Param>
        <Param id="$ParamName_02">$ParamContent_02</Param>
        <Param id="$ParamName_03">$ParamContent_03</Param>
        ...
    </Parameters>
</WSM::$Request >
```

The Product Catalogue (ProductID) defines the valid values for $ProductID, $Scope, and $ParamName. The Product Catalogue also defines the values for $PlanID in relation to the pseudo ProductIDs DIR (Directory Service), TCAS (Transaction Control and Status service) and OPER (Operational / runtime functions service), but the $PlanID values for real end user products such as an ADSL service are generally specific to each WSM client and are obtained from VOCUS during the implementation phase.

The data types definition document defines the permitted values for $ParamContent.

A WSM Soap Response follows a similar format to the request:

SOAP Response

```
<WSM::$Response>
    <TransactionID>$TransactionID</TransactionID>
    <ResponseType>$ResponseType</ResponseType>
    <Parameters>
        <Param id="$ParamName_01">$ParamContent_01</Param>
        <Param id="$ParamName_02">$ParamContent_02</Param>
        <Param id="$ParamName_03">$ParamContent_03</Param>
        <Param id="$ComplexRecordName_01">
            <![CDATA[
                <ComplexRecordName_01>
                    <ParamName_04>$ParamContent_04</ParamName_04>
                    <ParamName_05>$ParamContent_05</ParamName_05>
                    <ParamName_06>$ParamContent_06</ParamName_06>
                    <AuthResult>PASS</AuthResult>
                    <Reason>Accepted</Reason>
                </ComplexRecordName_01>
            ]]>
        </Param>
        ...
    </Parameters>
</WSM::$Response>
```

1.2 Synchronous and Asynchronous Response

The ResponseType tells the client how a transaction result will be delivered. If the value of ResponseType is SYNC, this means this is a synchronous transaction and the result will be available immediately. The WSM specification will clearly state which operations are synchronous and which are asynchronous.

The following Soap Response shows a synchronous transaction response:

Synchronous SOAP Response

```
<WSM::GetResponse>
    <TransactionID>CA2AF31712</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="ServiceType">ADSL1</Param>
        <Param id="Exchange">BELCONNEN</Param>
        <Param id="CCA">CANBERRA</Param>
        <Param id="State">ACT</Param>
        <Param id="Zone">METRO</Param>
    </Parameters>
</WSM::GetResponse>
```

If the value of $ResponseType is ASYNC, this means this is a synchronous transaction and the result will be available via call-back or by querying TCAS.

The following Soap Response shows an asynchronous transaction response. Notice that the 'Parameter' element is missing because it will be delivered (or retrieved) asynchronously:

Asynchronous SOAP Response

```
<WSM::GetResponse>
    <TransactionID>CA2AF31712</TransactionID>
    <ResponseType>ASYNC</ResponseType>
</WSM::GetResponse>
```

1.3 Obtaining an Asynchronous Response Result

There are two options for obtaining the result of an ASYNC transaction:

- Implement a SOAP call-back server.
- Implement a polling mechanism that will query the TCAS product/service for results

1.3.1 Implement a Call-Back Server

If the client chooses to use a call-back server, the WSM will perform a call-back to the client's server to deliver the summarised transaction result. A single callback operation may return the results for one or more transactions as illustrated below:

SOAP Call Back

```
<WSM::CallBackRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id="SUCCESS">BA2AF31871</Param>
        <Param id="SUCCESS">AA2AFE1571</Param>
        <Param id="SUCCESS">AC2AFE1873</Param>
        <Param id="SUCCESS">AEFFGE1812</Param>
```

```
        <Param id="SUCCESS">HHEFG918AA</Param>
    </Parameters>
</WSM::CallBackRequest>
```

After the call-back, the client can query TCAS to get the detail of the transaction – if required. Most of the time client will only need to do this for transactions that have the status FAILED.

The call-back mechanism may provide multiple updates for a transaction as it passes through various states in its lifecycle. See the TCAS Product documentation for further details about transaction states and transitions.

1.3.2 Implement a Polling Mechanism

If a client chooses to implement a polling mechanism, they will have to query TCAS as per the soap request below:

SOAP Request

```
<WSM::GetRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <ProductID>TCAS</ProductID>
    <PlanID>TRANSACTION-ID </PlanID>
    <Scope>RESPONSE</Scope>
    <Parameters>
        <Param id="TransactionID">BA2AF31871</Param>
    </Parameters>
</WSM::GetRequest>
```

TCAS will respond with the status or values of the transaction. The following sample response shows a transaction still in a pending state.

SOAP Response

```
<WSM::GetResponse>
    <TransactionID>984F31712</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="TransactionState">QUEUED</Param>
    </Parameters>
</WSM::GetResponse>
```

1.4 SOAP Faults and Failed Transactions

All transaction failures will result in a SOAP Fault that can be delivered immediately (synchronous) or delivered when the transaction has been completed (asynchronously).

Synchronous SOAP faults are usually generated by a synchronous WSM transaction or by data validation errors. The client will receive this type of SOAP fault immediately. The WSM uses a standardised SOAP fault format as shown below:

SOAP Response (Fault)

```
<SOAP-ENV:Fault>
   <faultcode>SOAP-ENV</faultcode>
   <faultstring>Certificate request rate limit exceeded<faultstring>
   <detail>
      <ns1:Fault>
         <TransactionID>984F31712</TransactionID>
```

```
          <ErrorCode>WSM-4080</ErrorCode>
      <ns1:Fault>
    </detail>
<SOAP-ENV:/Fault>
```

An asynchronous SOAP fault is usually generated by an asynchronous transaction. The details of the SOAP fault can be retrieved by querying the TransactionID using TCAS:

SOAP Request

```
<WSM::GetRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <ProductID>TCAS</ProductID>
    <PlanID>TRANSACTION-ID </PlanID>
    <Scope>RESPONSE</Scope>
    <Parameters>
        <Param id="TransactionID">A33AF31483</Param>
    </Parameters>
</WSM::GetRequest>
```

The response will be as follows:

SOAP Response

```
<WSM::GetResponse>
    <TransactionID>AABF31772</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="TransactionState">FAILED</Param>
        <Param id="faultcode">ADSL-2113</Param>
        <Param id="faultmessage">Service is suspended</Param>
    </Parameters>
</WSM::GetResponse>
```

1.5 Auditing and Tracing Using TCAS

All transactions can be audited using TCAS. For example, to trace the original request of the SOAP response in the preceding example:

SOAP Request

```
<WSM::GetRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <ProductID>TCAS</ProductID>
    <PlanID>TRANSACTION-ID</PlanID>
    <Scope>REQUEST</Scope>
    <Parameters>
        <Param id="TransactionID">A33AF31483</Param>
    </Parameters>
</WSM::GetRequest>
```

TCAS will respond with the original Soap Request content. From the ErrorCode and the Soap Request we can conclude that the client is trying to suspend a service that is already suspended.

SOAP Response

```
<WSM::GetResponse>
    <TransactionID>9988ABFE</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
```

```
        <Param id="Operation">SET</Param>
        <Param id="AccessKey">AABBCCDDFFEEGGHH</Param>
        <Param id="ProductID">ADSL</Param>
        <Param id="PlanID">HOSTED-STANDARD-40GB</Param>
        <Param id="Scope">STATUS</Param>
        <Param id="ServiceID">TEST-123</Param>
        <Param id="ServiceStatus">SUSPEND</Param>
    </Parameters>
</WSM::GetResponse>
```

The client can also use TCAS to trace the steps of a transaction as shown in the following sample Soap Request:

SOAP Request

```
<WSM::GetRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <ProductID>TCAS</ProductID>
    <PlanID>TRANSACTION-ID</PlanID>
    <Scope>STEPS</Scope>
    <Parameters>
        <Param id="TransactionID">A33AF31483</Param>
    </Parameters>
</WSM::GetRequest>
```

TCAS will respond synchronously as shown below. The transaction was FAILED at service status validation. This caused processing to jump to the "Callback" step.

SOAP Response

```
<WSM::GetResponse>
    <TransactionID>8988ACDE</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="SUCCESS">Validate XML</Param>
        <Param id="SUCCESS">AccessKey Permission Validation</Param>
        <Param id="SUCCESS">Check ServiceID ownership</Param>
        <Param id="FAILED">Lookup current service status</Param>
        <Param id="SUCCESS">Callback</Param>
    </Parameters>
</WSM::GetResponse>
```

2 WSM Session Examples

The client's web services application will login using HTTP Client Certificate Authentication. This is also known as 2 way SSL. WSM accepts authentication certificates in keystore format.

2.1 Login to WSM

The certificate login sequence is as follow:

1. The client application sends the authentication certificate to the WSM login server.
   The URL for certificate authentication is: **https://wsm.webservice.m2.com.au:9443/login/**
2. The WSM login server confirms the validity of the certificate and creates a session. HTTP status code **202 ACCEPTED** will be returned to the client.
3. The client disconnects from the WSM login server.

Following a successful certificate authentication request, WSM will create a transaction session that has an idle timeout of 600 seconds. The client will then send SOAP requests for that session to the URL specified in the WSDL file.

2.2 Maintaining a WSM Session

Once client session is created, WSM will maintain the session with an idle timeout of 600 seconds. Any request sent to WSM will maintain the session for another 600 seconds.

2.3 Logout from WSM

To logout from WSM, simply leave the session idle for more than 600 seconds.

3 ProductID ADSL Examples

This documentation contains examples for ProductID ADSL and its interoperability with other WSM ProductIDs. These example SOAP XML messages have been simplified to improve readability.

The ProductID ADSL examples will show many scenarios which involve the following ProductID:

ProductID	PlanID	Description
ADSL	HOSTED-STANDARD-40GB HOSTED-STANDARD-80GB	These are fictitious example PlanIDs which symbolise two ADSL plans with 40 & 80 GB monthly usage allowances respectively, on standard shaping. Refer to the PlanID example in this document for the plan characteristics.
TCAS	TRANSACTION-ID	This is a hardcoded PlanID from TCAS that we use for managing transactions.

These examples will show the timeline and lifecycle of an ADSL product from provisioning to cancellation. We will assume the Client deployed a working SOAP call-back server to receive all final transaction results.

The following chronological scenario will be covered by the examples:

1. Perform Service Qualification
2. Perform Username Search
3. Creation of an ADSL service
4. Receiving SOAP call-back
5. Getting the details of the ADSL service
6. Suspending & Resuming the ADSL service
7. Performing an update of multiple service parameters in a single transaction
8. Reading the TCAS transaction steps
9. Deactivation of ADSL services

3.1 Performing ADSL Service Qualification

Prior to provisioning it is advisable to perform a Service Qualification (SQ) to ensure the FNN is valid and the telephone exchange has been enabled for the desired broadband service. ***This activity is not required if the SQ and Carrier Provisioning activities are performed outside the scope of WSM. In this case the PlanID will indicate to WSM that it should assume successful carrier provisioning has occurred (if relevant) and the provisioning activity should only affect the internal OSS configuration.***

SOAP Request for a SQ

```
<WSM::GetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Scope>QUALIFY</Scope>
    <Parameters>
        <Param id="FNN">0396746494</Param>
    </Parameters>
</WSM::GetRequest>
```

The following synchronous response shows that the exchange is not enabled for the ADSL service.

SOAP Response for a Failed SQ

```
<SOAP-ENV:Fault>
   <faultcode>SOAP-ENV</faultcode>
   <faultstring>Exchange not enabled<faultstring>
   <detail>
      <ns1:Fault>
         <TransactionID>984F31712</TransactionID>
         <ErrorCode>ADSL-1300</ErrorCode>
      <ns1:Fault>
    </detail>
<SOAP-ENV:/Fault>
```

The following response shows a SQ that has been successful, so the client can proceed with provisioning of the service.

SOAP Response for a Successful SQ

```
<WSM::GetResponse>
    <TransactionID>984F31712</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="ServiceType">"ADSL1","ADSL2"</Param>
        <Param id="Exchange">LONSDALE</Param>
        <Param id="CCA">MELBOURNE</Param>
        <Param id="State">VIC</Param>
        <Param id="Zone">METRO</Param>
    </Parameters>
</WSM::GetResponse>
```

3.2 Performing a Search by Username

This function can be used by the client prior to submitting the provisioning request for a new service or change of username to ensure the requested username is not already in use by another service. This is not necessary if the client is already tracking usage of usernames. The request uses scope SEARCH and the username includes the authentication realm as described in the WSM data types

definition document. The service status can optionally be included to filter the responses to one category of status.

Search by Username SOAP Request

```
<WSM::GetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Scope>SEARCH</Scope>
    <Parameters>
        <Param id="Username">johndoe@xyz.your-dsl.net</Param>
        <Param id="ServiceStatus">ACTIVE</Param>
    </Parameters>
</WSM::GetRequest>
```

If the username already exists on another service the synchronous response will return the ServiceID of that service:

Search by Username SOAP Response

```
<WSM::GetResponse>
    <TransactionID>99812BDEF</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="ServiceID">TEST-123</Param>
    </Parameters>
</WSM::GetResponse>
```

Otherwise an empty respond will be returned:

Search by Username SOAP Fault

```
<WSM::GetResponse>
    <TransactionID>99812BDEF</TransactionID>
    <ResponseType>SYNC</ResponseType>
</WSM::GetResponse>
```

3.3 Creation of an ADSL Service

After passing a SQ, the client can proceed with ADSL provisioning.

We will use PlanID HOSTED-STANDARD-40GB. This PlanID serves as a template for the provisioning activity. Please refer to the PlanID example chapter for the detailed datasheet.

Some of the important characteristics of this PlanID are as follow:

- Standard shaping with 40GB quota will be applied
- IP assignment is dynamic
- No billing or upstream carrier provisioning is required
- DIR entry is not required
- Postcode is being provided as LocationReference so the correct IP address range can be used when assigning the IP address.
- WSM will not expect a DirectoryID to be supplied for this service (details not required as carrier provisioning is performed outside WSM).

Create New Service SOAP Request

```
<WSM::CreateRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="Username">reddit@xyz.your-dsl.net</Param>
        <Param id="Password">password123</Param>
        <Param id="CarrierID">0396420560</Param>
        <Param id="AnniversaryDay">1</Param>
        <Param id="LocationReference">3006</Param>
    </Parameters>
</WSM::CreateRequest>
```

WSM will accept the transaction, placing it in the asynchronous transaction queue and returning the following Soap response:

Create New Service SOAP Response

```
<WSM::CreateResponse>
    <TransactionID>EF99182AC</TransactionID>
    <ResponseType>ASYNC</ResponseType>
</WSM::GetResponse>
```

The transaction is now entering its life cycle and will be processed according to the business rules of the PlanID.

3.4 Receiving Soap Call-Back

Once the transaction is complete, WSM will perform a call-back to the client's SOAP server with the result:

Create New Service SOAP Call-Back

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id="SUCCESS">EF99182AC </Param>
    </Parameters>
</WSM::CallBack>
```

Transaction has successfully completed (status of SUCCESS). The service has been provisioned in VOCUS's hosted services infrastructure.

3.5 Getting Details of a Service

The client can retrieve the details of a service using the GET operation with scope SERVICE. Information regarding the IP address can also be retrieved using this scope.

Get Service Details SOAP Request

```
<WSM::GetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <Scope>SERVICE</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
    </Parameters>
</WSM::GetRequest>
```

The synchronous response will return the service details. Notice that the IP address value is 0.0.0.0 because the IP assignment is DYNAMIC. To check the dynamically assigned IP address for a current online session the client should use WSM:Get with ProductID OPER and Scope ONLINE-SESSION.

Get Service Details SOAP Response

```
<WSM::GetResponse>
    <TransactionID>HG00199AF</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="Username">reddit@xyz.your-dsl.net</Param>
        <Param id="Password">password123</Param>
        <Param id="CarrierID">0396420560</Param>
        <Param id="ServiceStatus">ACTIVE</Param>
        <Param id="ShapeStatus">FALSE</Param>
        <Param id="PlanID">HOSTED-STANDARD-40GB</Param>
        <Param id="AnniversaryDay">1</Param>
        <Param id="IPAddress">0.0.0.0</Param>
        <Param id="IPAssignment">DYNAMIC </Param>
        <Param id="LocationReference">3006</Param>
    </Parameters>
</WSM::GetResponse>
```

3.6 Suspending and Resuming the ADSL Service

The following examples illustrate the operations used to suspend and then resume the ADSL service.

These operations are performed using operation SET with scope STATUS.

To SUSPEND a service:

Suspend Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="ServiceStatus">SUSPEND</Param>
    </Parameters>
</WSM::SetRequest>
```

The final result will be delivered via Soap call-back:

Suspend Service SOAP Response

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id="SUCCESS">AEF8991DD</Param>
    </Parameters>
</WSM::CallBack>
```

Calling operation GET with scope SERVICE will provide the following response that clearly shows the service is suspended:

Get Service Details SOAP Response

```
<WSM::GetResponse>
    <TransactionID>HG00199AF</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="Username">reddit@xyz.your-dsl.net</Param>
        <Param id="Password">password123</Param>
        <Param id="CarrierID">0396420560</Param>
        <Param id="ServiceStatus">SUSPEND</Param>
        <Param id="ShapeStatus">TRUE</Param>
        <Param id="PlanID">HOSTED-STANDARD-40GB</Param>
        <Param id="AnniversaryDay">1</Param>
        <Param id="IPAddress">0.0.0.0</Param>
        <Param id="IPAssignment">DYNAMIC </Param>
        <Param id="LocationReference">3006</Param>
    </Parameters>
</WSM::GetResponse>
```

To resume the service, use operation SET with scope SERVICE and set the ServiceStatus param to value ACTIVE:

Resume Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="ServiceStatus">ACTIVE</Param>
    </Parameters>
</WSM::SetRequest>
```

The transaction result will be delivered by Soap call-back.

3.7 Performing an Update of Multiple Service Parameters in a Single Transaction

The following example modifies the Password, Carrier ID, Anniversary Day (Monthly allowance reset day) and Postcode of the ADSL Service.

Update Service Details SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="Password">password123</Param>
        <Param id="CarrierID">0396420569</Param>
        <Param id="AnniversaryDay">15</Param>
        <Param id="LocationReference">9999</Param>
    </Parameters>
</WSM::SetRequest>
```

This is an asynchronous operation so the results will be delivered via call-back. In this example the transaction did not finish successfully.

Update Service Details SOAP Call-Back

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id="FAILED">AEAEBB00D</Param>
    </Parameters>
</WSM::CallBack>
```

The client will now need to drill down into the transaction to see which param has failed. This is performed using TCAS operation GET scope STEPS as shown in the next section.

3.8 Reading TCAS Transaction Steps

Scope STEPS is used to check at which transaction step a multiple update operation failed

TCAS SOAP Request

```
<WSM::GetRequest>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <ProductID>TCAS</ProductID>
    <PlanID>TRANSACTION-ID</PlanID>
    <Scope>STEPS</Scope>
    <Parameters>
        <Param id="TransactionID">AEAEBB00D</Param>
    </Parameters>
</WSM::GetRequest>
```

TCAS will synchronously send a response as shown below. The transaction FAILED at LocationReference (postcode) validation. All other details have been updated successfully.

TCAS SOAP Response

```
<WSM::GetResponse>
    <TransactionID>8988ACDE</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="SUCCESS">Validate XML</Param>
        <Param id="SUCCESS">AccessKey Permission Validation</Param>
        <Param id="SUCCESS">Check ServiceID ownership</Param>
        <Param id="SUCCESS">Update service password</Param>
        <Param id="SUCCESS">Update Carrier ID</Param>
        <Param id="SUCCESS">Update Service Anniversary Day</Param>
        <Param id="FAILED">Update LocationReference</Param>
        <Param id="SUCCESS">Callback</Param>
    </Parameters>
</WSM::GetResponse>
```

The LocationReference '9999' that the client sent is not a valid postcode. The client now has the option of correcting the transaction by resending with correct postcode:

Update Service Details SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="LocationReference">3009</Param>
    </Parameters>
```

```
</WSM::SetRequest>
```

3.9 Deactivation of ADSL Service

The last example illustrates how the client can cancel the services. This is done by simply updating the ServiceStatus for each service to INACTIVE:

Cancel Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>ADSL</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="ServiceStatus">INACTIVE</Param>
    </Parameters>
</WSM::SetRequest>
```

The final result will be delivered via Soap call-back.

4 ProductID FIBRE Examples

This documentation contains examples for ProductID FIBRE and its interoperability with other WSM ProductIDs. These example SOAP XML messages have been simplified to improve readability.

NBN provisioning is complex with multiple different flows and permutations. This documentation will not be able detail each one of these flows and permutations. The purpose of this example is to explain at high level WSM web service works for ProductID FIBRE.

The ProductID FIBRE examples will show many scenarios which involve the following ProductID:

ProductID	PlanID	Scope	Description
FIBRE	RC-DEFAULT-MAP NCL2-DSCP-MAP	RESELLER-CONNECT NETWORK-CONNECT	These are fictitious example PlanIDs which symbolise two FIBRE plans of Reseller-Connect with default map profile; And Network-Connect Layer 2 with DSCP map
DIR	BROADBAND	-	This is a hardcoded PlanID from DIR that we use for managing NBN Address ID
TCAS	TRANSACTION-ID	-	This is a hardcoded PlanID from TCAS that we use for managing transactions.
CPE	-	SERVICE	Retrieve the detail of CPE service

These examples will show the lifecycle of an FIBRE product from provisioning. We will also include a CPE in this order.

We will cover 2 separate scenarios:

1. Ordering of a brand-new Reseller-Connect Layer 3 tail
2. Ordering of a transfer Network-Connect Layer 2 tail

The following chronological scenario will be covered by the examples for Reseller-Connect tail:

1. Perform Address Lookup
2. Perform Service Qualification
3. Create address entry for CPE and NCD order
4. Creation of a **New** FIBRE Reseller-Connect service
5. Pooling of FIBRE order
    a. Checking overall order progress
    b. Checking appointment progress
    c. Checking NBN notifications
6. Getting the details of the CPE service
7. Getting the details of the FIBRE service

The following chronological scenario will be covered by the examples for Network-Connect tail:

1. Perform Address Lookup
2. Perform Service Qualification
3. Creation of a FIBRE Network-Connect L2 service by means of **Transfers**
4. Pooling of FIBRE order
    a. Checking overall order progress
    b. Checking NBN order & appointment notifications
5. Getting the details of the FIBRE service

4.1 Ordering of New RESELLER-CONNECT Tail as NEW Order

In these examples, we will be ordering a brand-new Reseller-Connect service as a NEW order. This is Vocus Wholesale NBN Layer 3 (hosted) product.

Note – For an example to order a TRANSFERS, please refer to NETWORK-CONNECT examples at section 4.2. Whilst the product is different, the mechanism is the same.

This example order will be consisting of:

- An NBN tail – We will order an FTTC service
- A Vocus issued CPE NBN modem
- An NBN issued NCD for FTTC

Each NBN technology type have their own specific ordering characteristic. This influence the WSM request params need to be sent on an order. ProductID FIBRE specification will details which params are specific for a certain technology type.

4.1.1 Performing Address Lookup

Prior to service qualification, it is necessary to perform an address lookup to retrieve the exact Vocus Address ID for destination service address.

This is performed via ProductID DIR::Get with PlanID BROADBAND

SOAP Request for an Address Lookup

```
<WSM:GetRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>DIR</ProductID>
   <PlanID>BROADBAND</PlanID>
   <Parameters>
      <Param id="Main.StreetName">Example</Param>
      <Param id="Main.StreetType">St</Param>
      <Param id="Main.Suburb">Upper Melbourne</Param>
      <Param id="Main.State">QLD</Param>
      <Param id="Main.Postcode">4009</Param>
      <Param id="SearchMode">DEFAULT</Param>
   </Parameters>
</WSM:GetRequest>
```

The following synchronous response shows returns multiple address matches. The operator will need to choose the correct address Directory ID.

SOAP Response for a Successful Address Lookup

```
<WSM:GetResponse>
   <TransactionID>1686B5F63</TransactionID>
   <ResponseType>SYNC</ResponseType>
   <Parameters>
      <Param id="FibreAddressRecord"><![CDATA[<BroadbandAddressRecord><DirectoryID>
LOC000340636851</DirectoryID><Carrier>NBN</Carrier><AddressLong>UNIT 1 99 EXAMPLE ST
UPPER MELBOURNE QLD</AddressLong></BroadbandAddressRecord>]]></Param>
      <Param id="FibreAddressRecord"><![CDATA[<BroadbandAddressRecord><DirectoryID>
LOC000340636852</DirectoryID><Carrier>NBN</Carrier><AddressLong>UNIT 2 99 EXAMPLE ST
UPPER MELBOURNE QLD</AddressLong></BroadbandAddressRecord>]]></Param>
      <Param id="FibreAddressRecord"><![CDATA[<BroadbandAddressRecord><DirectoryID>
LOC000340636853</DirectoryID><Carrier>NBN</Carrier><AddressLong>UNIT 3 99 EXAMPLE ST
UPPER MELBOURNE QLD</AddressLong></BroadbandAddressRecord>]]></Param>
```

```
    </Parameters>
</WSM:GetResponse>
```

Let assume operator has chosen the first entry with Vocus Address ID `LOC000340636851`. We will use this Address ID for the next step, performing service qualification.

4.1.2 Performing Service Qualification

Prior to provisioning it is required to perform a Service Qualification (SQ) to ensure the address has been enabled for the desired broadband service and to retrieve several important information to be passed on the actual order.

Operator has chosen the first entry with Vocus Address ID `LOC000340636851` to perform service qualification. SQ is performed via ProductID FIBRE::Get scope QUALIFY.

SOAP Request for a SQ

```
<WSM:GetRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>FIBRE</ProductID>
   <Scope>QUALIFY</Scope>
   <Parameters>
      <Param id="DirectoryID">LOC000340636851</Param>
   </Parameters>
</WSM:GetRequest>
```

The following response shows a SQ that has been successful with a PASS result, so client can proceed with ordering of the service.

The below result tells us this is a FTTC Service Class 33 address. For class 33, we have the option for end user NCD Self Install. This option will require an NCD hardware shipment from Vocus warehouse.

SOAP Response for a Successful SQ

```
<WSM:GetResponse>
   <TransactionID>1686BBDD</TransactionID>
   <ResponseType>SYNC</ResponseType>
   <Parameters>
   <Param id="Result">PASS</Param>
   <Param id="ServiceType">FTTC</Param>
   <Param id="ServiceClass">32</Param>
   <Param id="ConnectionType">Type 1</Param>
   <Param id="CopperDisconnectionDate">20210212</Param>
   <Param id="Zone">Urban</Param>
   <Param id="DevelopmentCharge">FALSE</Param>
   <Param id="CSA">CSA600000000099</Param>
   <Param id="CVCID">Auto-Assigned</Param>
   <Param id="CopperPairRecord"><![CDATA[
      <CopperPairRecord>
        <CopperPairID>CPI300010022999</CopperPairID>
        <CopperPairStatus>Inactive</CopperPairStatus>
        <NBNServiceStatus>Line In Use</NBNServiceStatus>
        <ServiceClass>32</ServiceClass>
        <POTSInterconnect>N/A</POTSInterconnect>
        <POTSMatch>FALSE</POTSMatch>
        <TC2Speed>5Mbps,10Mbps,20Mbps</TC2Speed>
        <UploadSpeed>20-40</UploadSpeed>
        <DownloadSpeed>50-100</DownloadSpeed>
        <NetworkCoExist>TRUE</NetworkCoExist>
        <ExtraCharge>FALSE</ExtraCharge>
      </CopperPairRecord>
   ]]></Param>
   </Parameters>
```

```
</WSM:GetResponse>
```

Different technology type will return different result. For example for FTTP, we will get `NBNPortRecords` instead of `CopperPairRecords`.

4.1.3 Create address entry for CPE and NCD order

For FTTC class 33, we can choose to have the NCD shipped from Vocus warehouse. We will also order a Vocus issued NBN Modem as part of this order.

To be able to do this, we would need to create an address entry in WSM and pass the created Address ID when we order the NBN (FIBRE) service.

SOAP Request to Create and Address Record

```
<WSM::CreateRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>DIR</ProductID>
   <PlanID>STANDARD</PlanID>
   <Parameters>
      <Param id="FirstName">Yogesh</Param>
      <Param id="LastName">Shinde</Param>
      <Param id="DOB">19770101</Param>
      <Param id="Main.AddressType">UNT</Param>
      <Param id="Main.StreetName">WILLIAM</Param>
      <Param id="Main.StreetType">ST</Param>
      <Param id="Main.Street1stNumber">369</Param>
      <Param id="Main.Unit1stNumber">2</Param>
      <Param id="Main.Suburb">MELBOURNE</Param>
      <Param id="Main.State">VIC</Param>
      <Param id="Main.PostCode">3000</Param>
      <Param id="Main.Country">61</Param>
      <Param id="Normal.ContactFirstName">Yogesh</Param>
      <Param id="Normal.ContactLastName">Shinde</Param>
      <Param id="Normal.Phone">0433999777</Param>
      <Param id="Emergency.ContactFirstName">John </Param>
      <Param id="Emergency.ContactLastName">Doe</Param>
      <Param id="Emergency.Phone">0433111222</Param>
   </Parameters>
</WSM::CreateRequest>
```

A Vocus Address ID with ID `FFF6A2EBBC31021` is returned:

SOAP Response for a Successful Address creation

```
<WSM:CreateResponse>
   <TransactionID>1686BBDD</TransactionID>
   <ResponseType>SYNC</ResponseType>
   <Parameters>
      <Param id="DirectoryID">FFF6A2EBBC31021</Param>
   </Parameters>
</WSM:GetResponse>
```

4.1.4 Creation of a FIBRE Reseller-Connect service

FIBRE::Get scope QUALIFY above has given a PASS result. We can now proceed with calling FIBRE::Create with scope RESELLER-CONNECT to start the provisioning flow for FTTC service.

We will all the previous value retrieved from previous method: `LOC000340636851, DirectoryID FFF6A2EBBC31021, PlanID RC-DEFAULT-MAP, PlanID for NBN Modem MY-MODEM-PLAN, Copper Pair ID CPI300010022999.` And PlanID for `NBNCPEPlanID FTTCNCDRC`

Valid value for `NBNCPEPlanId` is available at ProductID FIBRE appendix. We have chosen `FTTCNCDRC` since this will enable NCD shipment from Vocus warehouse to CPEDirectoryID `FFF6A2EBBC31021`.

FIBRE PlanID and CPEPlanID will be provided as part of B2B onboarding.

SOAP Request to Create a NBN FTTC service as NEW order

```
<WSM::CreateRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>FIBRE</ProductID>
   <PlanID> RC-DEFAULT-MAP</PlanID>
   <Scope>RESELLER-CONNECT</Scope>
   <Parameters>
      <Param id="ServiceID">MY-SERVICE-ID</Param>
      <Param id="Password">Kl*&AdacnZ!d$</Param>
      <Param id="CustomerName">Yogesh Shinde</Param>
      <Param id="Phone">0433111222</Param>
      <Param id="LocationReference">3000</Param>
      <Param id="DirectoryID">LOC000340636851</Param>
      <Param id="Realm">wba.samplewsm.com.au</Param>
      <Param id="ServiceLevel">STANDARD</Param>
      <Param id="ServiceType">FTTC</Param>
      <Param id="CopperPairID">CPI300010022999</Param>
      <Param id="CPEDirectoryID">FFF6A2EBBC31021</Param>
      <Param id="CPEPlanId">MY-MODEM-PLAN</Param>
      <Param id="NBNCPEPlanId">FTTCNCDRC</Param>
   </Parameters>
</WSM::CreateRequest>
```

The order is now entering its life cycle and will be processed according to the business rules of the PlanID.

Create New Service SOAP Response

```
<WSM::CreateResponse>
    <TransactionID>EF99182AC</TransactionID>
    <ResponseType>ASYNC</ResponseType>
</WSM::CreateResponse>
```

The method FIBRE::Create() is a ASYNC method. This method will initially returns a TransactionID `EF99182AC` with no other details. As the order progresses, client can pool TCAS::Get with scope RESPONSE to retrieve additional order details such as PRI, Appointments, and the final result of the order which can be either SUCCESS or FAILED.

WSM will immediately try to perform validation on the order. If there is any detectable invalid values, the call to FIBRE::Create will FAILED immediately as a SYNC transaction instead of ASYNC.

4.1.5 Receiving Soap Call-Back

Once the transaction is complete, WSM will perform a call-back to the client's SOAP server with the result:

Create New Service SOAP Call-Back

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id=”SUCCESS”>EF99182AC</Param>
    </Parameters>
</WSM::CallBack>
```

4.1.6 Pooling of FIBRE order

Call-back is the preference for getting completion of orders. However, WSM also provides a polling mechanism. Once NBN order received by WSM and a TransactionID is issued, client can start perform pooling to get order details. Or utilise WSM callback to receive the very final end status of orders.

Transaction will initially on QUEUED state. This means WSM has received and ready to submit to upstream carrier. Once upstream has received the order, the status will be PROCESSING. From this status, an order can either become SUCCESS or FAILED.

For NBN order there are 3 different pooling can be performed. And the recommended pooling frequency are as follow:

- Checking overall order progress – every 6 to 8 hours
- Checking NBN order & appointment notifications – once an hour

Excessive use of pooling will trigger automatic ban within WSM.

4.1.6.1 Checking Overall Order Progress

To check overall order progress, client will use ProductID TCAS::Get scope RESPONSE to check the progress of TransactionID `EF99182AC`

SOAP Request to Check Overall Order Progress

```
<WSM::GetRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>TCAS</ProductID>
   <PlanID>TRANSACTION-ID</PlanID>
   <Scope>RESPONSE </Scope>
   <Parameters>
      <Param id="TransactionID">EF99182AC</Param>
   </Parameters>
</WSM::GetRequest>
```

Initially, no details will be returned:

SOAP Response of Checking Overall Order Progress

```
<WSM::GetResponse>
    <TransactionID>ASDFGHJI</TransactionID>
    <ResponseType>ASYNC</ResponseType>
   <Parameters>
      <Param id="TransactionState">Processing</Param>
      <Param id="BillingProviderID">12345678</Param>
   </Parameters>
</WSM::GetResponse>
```

As the order progresses further details will be returned as per below example.

This order include an FTTC NCD shipped from Vocus warehouse. For such order `NBNCPEServiceID` will returned. Client can use this ServiceID to query ProductID CPE::Get scope SERVICE to retrieve details of this hardware, including the Postage Tracking Number.

SOAP Response of Checking Overall Order Progress

```
<WSM::GetResponse>
    <TransactionID>JKJDNZA</TransactionID>
    <ResponseType>ASYNC</ResponseType>
   <Parameters>
      <Param id="TransactionState">PROCESSING</Param>
      <Param id="BillingProviderID">12345678</Param>
      <Param id="AppointmentSlot">AM</Param>
      <Param id="StartDateTime">20191111090000</Param>
      <Param id="EndDateTime">20191111130000</Param>
      <Param id="NBNInstanceID">PRI000191999999</Param>
      <Param id="AVCID">AVC000099999999</Param>
      <Param id="CVCID">CVC000099999999</Param>
      <Param id="CPEServiceID">NBNC1568888</Param>
      <Param id="NBNCPEServiceID">NBNC15699999</Param>
   </Parameters>
</WSM::GetResponse>
```

And finally, when the order completed, the TransactionStatus will shown as SUCCESS (or FAILED).

Furthermore, certain ProductID TCAS::Get scope RESPONSE for NEW or TRANSFERS orders will return Appointment Details – AppointmentSlot, StartDateTime, and EndDateTime. These appointment can be further drilled down using FIBRE::Get scope NOTIFICATIONS.

Some order will also returns `NBNCPEServiceID` param. This is a WSM CPE ServiceID which can be queried to retrieve details of a CPE, such as shipment tracking number. This will be detailed further down in this documentation.

SOAP Response of Checking Overall Order Progress

```
<WSM::GetResponse>
    <TransactionID>ASDFGHJI</TransactionID>
    <ResponseType>ASYNC</ResponseType>
   <Parameters>
      <Param id="TransactionState">Processing</Param>
      <Param id="BillingProviderID">12345678</Param>
      <Param id="AppointmentSlot">AM</Param>
      <Param id="StartDateTime">20191111090000</Param>
      <Param id="EndDateTime">20191111130000</Param>
      <Param id="NBNInstanceID">PRI000191999999</Param>
      <Param id="AVCID">AVC000099999999</Param>
      <Param id="CVCID">CVC000099999999</Param>
      <Param id="CPEServiceID">NBNC1568888</Param>
      <Param id="NBNCPEServiceID">NBNC15699999</Param>
   </Parameters>
</WSM::GetResponse>
```

4.1.6.2 Checking NBN Notifications

Vocus also provided a more detailed progress update for NBN NEW and TRANSFERS orders. This update can be retrieved using FIBRE::Get scope NOTIFICATIONS.

This method provides up to date progress reports on appointments and orders. Client will receive a “bulk” update for all orders at once for a requested timeframe.

SOAP Request of Checking NBN Notifications

```
<WSM:GetRequest>
        <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
        <ProductID>FIBRE</ProductID>
        <Scope>NOTIFICATIONS</Scope>
        <Parameters>
                <stan:Param id="StartDateTime">20191120220000</stan:Param>
        </Parameters>
</WSM:GetRequest>
```

SOAP Response of Checking NBN Notifications

```
<WSM:GetResponse>
        <TransactionID>KKDFGHJI</TransactionID>
        <ResponseType>SYNC</ResponseType>
        <Parameters>
                <Param id="MasterConversationRecord">
        <![CDATA[<MasterConversationRecord><DateTime>20191120224740</DateTime><Transa
ctionSubState>InProgress - Pending</TransactionSubState><Message>More Time Accepted.
</Message></ConversationRecord>]]>
                </Param>
                <Param id="ConversationRecord">
        <![CDATA[<MasterConversationRecord><DateTime>20191104165835</DateTime><Transa
ctionSubState>InProgress</TransactionSubState><Message>Order
Accepted.</Message></MasterConversationRecord>]]>
                </Param>
                <Param id="LastRecordID">12345678</Param>
                <Param id="HasMore">FALSE</Param>
        </Parameters>
</WSM:GetResponse>
```

4.1.7 Getting the details of the CPE service

Some order TCAS::Get scope RESPONSE will return `NBNCPEServiceID` or `CPEServiceID` param. NBNCPEServiceID is generally for NBN CPE such as FTTC NCD. While CPEServiceID is for a device such as Routers.

Client can query these ServiceID(s) to retrieve information such as Shipment Tracking. This can be performed using ProductID CPE::Get scope SERVICE.

SOAP Request of Checking NBN Notifications

```
<WSM:GetRequest>
        <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
        <ProductID>CPE</ProductID>
        <Scope>SERVICE</Scope>
        <Parameters>
                <Param id="ServiceID">NBNC15699999</Param>
        </Parameters>
</WSM:GetRequest>
```

SOAP Response of Checking NBN Notifications

```
<WSM:GetResponse>
        <TransactionID>QZDFGHJI</TransactionID>
```

```
        <ResponseType>SYNC</ResponseType>
        <Parameters>
                <Param id="ServiceID">NBNC15699999</Param>
                <Param id="ServiceStatus">ACTIVE</Param>
                <Param id="PlanID">FTTCNCDRC</Param>
                <Param id="SerialNumber">190902199909999</Param>
                <Param id="CustomerName">Yogesh Shinde</Param>
                <Param id="Phone">0433111222</Param>
                <Param id="Address">UNT 2 369 WILLIAM ST</Param>
                <Param id="Suburb">MELBOURNE</Param>
                <Param id="State">VIC</Param>
                <Param id="PostCode">3000</Param>
                <Param id="SenderCode">M2WYOGESH</Param>
                <Param id="OrderReference">9701999</Param>
                <Param id="Reference">200732991</Param>
                <Param id="BundleReference">12345678</Param>
                <Param id="DispatchRequest">03/12/2019</Param>
                <Param id="DispatchResponse">07/12/2019</Param>
                <Param id="ShipmentCompany">Toll</Param>
                <Param id="Consignment">AWVQ60899944</Param>
        </Parameters>
</WSM:GetResponse>
```

4.1.8 Getting the details of the FIBRE service

To retrieve a detail of a FIBRE service, client can use FIBRE::Get scope SERVICE. This method is not for checking order progress.

SOAP Request of Getting Service Details

```
<WSM:GetRequest>
        <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
        <ProductID>FIBRE</ProductID>
        <Scope>SERVICE</Scope>
        <Parameters>
                <Param id="ServiceID">MY-SERVICE-ID</Param>
        </Parameters>
</WSM:GetRequest>
```

SOAP Response of Getting Service Details

```
<WSM:GetResponse>
        <TransactionID>ZADFGHJI</TransactionID>
        <ResponseType>SYNC</ResponseType>
        <Parameters>
                <Param id="ServiceID">MY-SERVICE-ID</Param>
                <Param id="ServiceScope">RESELLER-CONNECT</Param>
                <Param id="ServiceStatus">ACTIVE</Param>
                <Param id="PlanID">RC-DEFAULT-MAP</Param>
                <Param id="Username">000007788990011@wba.samplewsm.com.au</Param>
                <Param id="OrderType">NEW</Param>
                <Param id="ServiceType">FTTC</Param>
                <Param id="CustomerName">Yogesh Shinde</Param>
                <Param id="Phone">0433111222</Param>
                <Param id="CopperPairID">CPI300010022999</Param>
                <Param id="ServiceLevel">STANDARD</Param>
                <Param id="NBNInstanceID">PRI000191999999</Param>
                <Param id="DirectoryID">LOC000340636851</Param>
                <Param id="CPEDirectoryID">FFF6A2EBBC31021</Param>
                <Param id="NBNCPEServiceID">NBNC15699999</Param>
                <Param id="NBNCPEPlanID">FTTCNCDRC</Param>
                <Param id="LocationReference">3000</Param>
                <Param id="AVCID">AVC000099999999</Param>
                <Param id="AggregatedVirtualCircuit">AVC000099999999</Param>
                <Param id="CVCID">CVC000099999999</Param>
```

```
                <Param id="BillingProviderID">12345678</Param>
        </Parameters>
</WSM:GetResponse>
```

4.1.9 Suspending and Resuming the FIBRE Service

The following examples illustrate the operations used to suspend and then resume the FIBRE service. This operation only valid for Layer 3 / Reseller-Connect.

These operations are performed using operation SET with scope STATUS.

To SUSPEND a service:

Suspend Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>FIBRE</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">MY-SERVICE-ID</Param>
        <Param id="ServiceStatus">SUSPEND</Param>
    </Parameters>
</WSM::SetRequest>
```

The final result will be delivered via Soap call-back:

Suspend Service SOAP Response

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id="SUCCESS">AEF8991DD</Param>
    </Parameters>
</WSM::CallBack>
```

To resume the service, use operation SET with scope SERVICE and set the ServiceStatus param to value ACTIVE:

Resume Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>FIBRE</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">MY-SERVICE-ID</Param>
        <Param id="ServiceStatus">ACTIVE</Param>
    </Parameters>
</WSM::SetRequest>
```

The transaction result will be delivered by Soap call-back.

4.1.10 Deactivation of FIBRE Service

The last example illustrates how the client can cancel the services. This is done by simply updating the ServiceStatus for each service to INACTIVE:

Cancel Service SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>FIBRE</ProductID>
    <Scope>STATUS</Scope>
    <Parameters>
        <Param id="ServiceID">MY-SERVICE-ID</Param>
        <Param id="ServiceStatus">INACTIVE</Param>
    </Parameters>
</WSM::SetRequest>
```

The final result will be delivered via Soap call-back.

4.2 Ordering of New NETWORK-CONNECT Tail as TRANSFER Order

In these examples, we will be ordering a brand-new Network-Connect NCL2 service as a TRANSFER order. This is Vocus Wholesale NBN Layer 2 product.

We will be re-using and referring a lot to RESELLER-CONNECT examples at section 4.1. Whilst the product is different, the mechanism is the same. The biggest differences in ordering NCL2 service is the inclusion of Vocus AggregatedCVC.

This example order will be consisting of:

- An NBN tail – We will order an FTTC service
- An NBN issued NCD for FTTC

Each NBN technology type have their own specific ordering characteristic. This influence the WSM request params need to be sent on an order. ProductID FIBRE specification will details which params are specific for a certain technology type.

4.2.1 Performing Address Lookup

Address lookup will be the same as RESELLER-CONNECT.

4.2.2 Performing Service Qualification

Service Qualification will be similar with RESELLER-CONNECT example. The only difference is we will include the AggCVC ID

SOAP Request for a SQ

```
<WSM:GetRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>FIBRE</ProductID>
   <Scope>QUALIFY</Scope>
   <Parameters>
      <Param id="DirectoryID">LOC000340636851</Param>
      <Param id="AVCID">AVC000340636851</Param>
      <Param id="CVCID">AggCVC000000999</Param>
   </Parameters>
</WSM:GetRequest>
```

The following response shows a SQ that has been successful with a PASS result, so client can proceed with ordering of the service.

The below result tells us this is a FTTC Service Class 33 address. For class 33, we have the option for end user NCD Self Install. This option will require an NCD hardware shipment from Vocus warehouse.

SOAP Response for a Successful SQ

```
<WSM:GetResponse>
   <TransactionID>1686BBDD</TransactionID>
   <ResponseType>SYNC</ResponseType>
   <Parameters>
   <Param id="Result">PASS</Param>
   <Param id="ServiceType">FTTC</Param>
   <Param id="ServiceClass">32</Param>
   <Param id="ConnectionType">Type 1</Param>
   <Param id="AVCIDMatch">TRUE</Param>
   <Param id="AVCIDTarget">CPI300010022999</Param>
   <Param id="CurrentEPID">0562</Param>
   <Param id="PrimaryTechnology">FTTC</Param>
   <Param id="CopperDisconnectionDate">20210212</Param>
   <Param id="Zone">Urban</Param>
   <Param id="DevelopmentCharge">FALSE</Param>
   <Param id="CSA">CSA600000000099</Param>
   <Param id="CVCID"> AggCVC000000999</Param>
   <Param id="CopperPairRecord"><![CDATA[
      <CopperPairRecord>
       <CopperPairID>CPI300010022999</CopperPairID>
       <CopperPairStatus>Inactive</CopperPairStatus>
       <NBNServiceStatus>Line In Use</NBNServiceStatus>
       <ServiceClass>32</ServiceClass>
       <POTSInterconnect>N/A</POTSInterconnect>
       <POTSMatch>FALSE</POTSMatch>
       <TC2Speed>5Mbps,10Mbps,20Mbps</TC2Speed>
       <UploadSpeed>20-40</UploadSpeed>
       <DownloadSpeed>50-100</DownloadSpeed>
       <NetworkCoExist>TRUE</NetworkCoExist>
       <ExtraCharge>FALSE</ExtraCharge>
      </CopperPairRecord>
   ]]></Param>
   </Parameters>
</WSM:GetResponse>
```

4.2.3 Create address entry for NCD order

This step will be exactly the same as RESELLER-CONNECT. Please refer to previous examples.

4.2.4 Creation of a FIBRE Network-Connect (NCL2) Service via TRANSFER

This step also similar with RESELLER-CONNECT. Client need to add AggCVC ID. In the example below, client did not specify a CTAG. Vocus will auto assign the CTAG.

Since this is a TRANSFERS order, client need to specify the OrderType as TRANSFER and also the CopperPairID to move.

SOAP Request to Create an NBN FTTC service as TRANSFER order

```
<WSM::CreateRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
```

```
    <ProductID>FIBRE</ProductID>
    <PlanID> NCL2-DSCP-MAP</PlanID>
    <Scope>RESELLER-CONNECT</Scope>
    <Parameters>
       <Param id="ServiceID">MY-SERVICE-ID</Param>
       <Param id="OrderType">TRANSFER</Param>
       <Param id="CustomerName">Yogesh Shinde</Param>
       <Param id="Phone">0433111222</Param>
       <Param id="CVCID">AggCVC000000999</Param>
       <Param id="LocationReference">3000</Param>
       <Param id="DirectoryID">LOC000340636851</Param>
       <Param id="AVCID">AVC000340636851</Param>
       <Param id="Realm">wba.samplewsm.com.au</Param>
       <Param id="ServiceLevel">STANDARD</Param>
       <Param id="ServiceType">FTTC</Param>
       <Param id="CopperPairID">CPI300010022999</Param>
       <Param id="CADate">20191110</Param>
    </Parameters>
</WSM::CreateRequest>
```

The order is now entering its life cycle and will be processed according to the business rules of the PlanID.

Create New Service SOAP Response

```
<WSM::CreateResponse>
    <TransactionID>AG99182AC</TransactionID>
    <ResponseType>ASYNC</ResponseType>
</WSM::CreateResponse>
```

4.2.5 Pooling of FIBRE NETWORK-CONNECT order

Pooling for NETWORK-CONNECT order is exactly the same with RESELLER-CONNECT. Please refers to RESELLER-CONNECT examples.

Below is an example of response for TCAS::Get scope RESPONSE showing CTAG which has been auto assigned.

The recommended interval for calling TCAS::Get scope RESPONSE is every 6 to 8 hours.

SOAP Response of Checking Overall Order Progress

```
<WSM::GetResponse>
    <TransactionID>ASDFGHJI</TransactionID>
    <ResponseType>ASYNC</ResponseType>
   <Parameters>
      <Param id="TransactionState">Processing</Param>
      <Param id="BillingProviderID">12345678</Param>
      <Param id="NBNInstanceID">PRI000191999999</Param>
      <Param id="AVCID">AVC000099999999</Param>
      <Param id="CVCID">AggCVC000000999</Param>
      <Param id="NBNCPEServiceID">NBNC15699999</Param>
      <Param id="CTAG">12345</Param>
   </Parameters>
</WSM::GetResponse>
```

Further pooling for NBN appointment and order notification are also supported for NETWORK-CONNECT. And the recommended pooling frequency are as follow:

- Checking NBN notifications – once an hour

4.3 FIBRE Events via Notification Mechanism

In these examples, we will be processing an event notification from WSM ProductID FIBRE for an NBN unplanned outage event. This notification are unsolicited events and initiated by Vocus.

Only client that subscribes to WSM call-back will receive these events notification.

4.3.1 Receiving Soap Event Notification

When an NBN unplanned outage occurred, WSM will perform an event notification in the same structure as call-back:

Receive New Service SOAP Event Notification

```
<WSM::CallBack>
    <AccessKey>AABBCCDDFFEEGGHH</AccessKey>
    <Parameters>
        <Param id=”EVENT”>EEAABBCCDD</Param>
    </Parameters>
</WSM::CallBack>
```

4.3.2 Querying Detail of Event Notification

Once an event notification is received, client can utilise ProductID TCAS operation GET with scope RESPONSE to query the event detail of TransactionID `EEAABBCCDD`

SOAP Request to Check Overall Order Progress

```
<WSM::GetRequest>
   <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
   <ProductID>TCAS</ProductID>
   <PlanID>TRANSACTION-ID</PlanID>
   <Scope>RESPONSE </Scope>
   <Parameters>
      <Param id="TransactionID">EEAABBCCDD</Param>
   </Parameters>
</WSM::GetRequest>
```

The NBN unplanned outage will be returned:

SOAP Response of Checking Overall Order Progress

```
<WSM::GetResponse>
    <TransactionID>ASDFGHJI</TransactionID>
<ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="TransactionState">PROCESSING</Param>
        <Param id="EventType">NBN-UNPLANNED-OUTAGE</Param>
        <Param id="EventID">INC000020057732</Param>
        <Param id="Priority">Medium</Param>
        <Param id="Status">Update</Param>
        <Param id="StartDateTime">20241202080000</Param>
        <Param id="Data">UnplannedOutage/INC000020057732-2024-08-23-1000.csv</Param>
        <Param id="Description">nbn recently resolved an unplanned outage in your
area. Resetting your home modem may optimise your nbn connection, otherwise reach out
```

```
to your service provider if you are still experiencing issues.</Param>
        <Param id="Notes">20241202090000: 01 The field technician is estimated to
arrive to the impacted site by 28/11/2024 by 08:30.</Param>
        <Param id="Notes">20241202100000: 02 The field technician is estimated to
arrive to the impacted site by 28/11/2024 by 08:30.</Param>
        <Param id="Notes">20241202110000: 03 The field technician is estimated to
arrive to the impacted site by 28/11/2024 by 08:30.</Param>
      </Parameters>
    </Parameters>
</WSM::GetResponse>
```

5 ProductID OPER Examples

The ProductID OPER examples illustrate several scenarios which involve the following ProductIDs:

ProductID	PlanID	Description
OPER	HOSTED-STANDARD-40GB	All example will be using ADSL PlanID
TCAS	TRANSACTION-ID	This is a hard-coded PlanID from TCAS that we use for managing transactions.

We will assume the Client has created a session with WSM, have deployed a working SOAP call-back server to receive all final transaction results, and have an active ADSL service in database.

The following scenarios will be covered by the examples:

- Getting the authentication log
- Disconnect an online Service

5.1 Getting Authentication History for a Service

Use operation GET with scope AUTH-LOG to retrieve the historical authentication attempt details for a service. The operation can only return authentication attempts that occurred within the previous 48 hours:

Authentication Attempt History SOAP Request

```
<WSM::GetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>OPER</ProductID>
    <PlanID>AUTH-LOG</PlanID>
    <Scope>USAGE-DAILY</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="ProductID">ADSL</Param>
    </Parameters>
</WSM::GetRequest>
```

The following response shows the service has tried to authenticate twice in the last 48 hours, both times successfully:

Authentication Attempt History SOAP Response

```
<WSM::GetResponse>
    <TransactionID>984F31712</TransactionID>
    <ResponseType>SYNC</ResponseType>
    <Parameters>
        <Param id="AuthUsageRecord">
            <![CDATA[
                <AuthUsageRecord>
                    <DateTime>20100725130000</DateTime>
                    <AuthResult>PASS</AuthResult>
                    <Reason>Accepted</Reason>
                </AuthUsageRecord>
            ]]>
        </Param>
        <Param id="AuthUsageRecord">
            <![CDATA[
                <AuthUsageRecord>
```

```
                    <DateTime>20100724123000</DateTime>
                    <AuthResult>PASS</AuthResult>
                    <Reason>Accepted</Reason>
                </AuthUsageRecord>
            ]]>
        </Param>
    </Parameters>
</WSM::GetResponse>
```

5.2 Disconnect an Online Session

For support or troubleshooting purposes it is often necessary to disconnect a current online session. The following example illustrates this type of request:

Disconnect Session SOAP Request

```
<WSM::SetRequest>
    <AccessKey>ABCDEFGHIJKLMNOP</AccessKey>
    <ProductID>OPER</ProductID>
    <PlanID>HOSTED-STANDARD-40GB</PlanID>
    <Scope>DISCONNECT</Scope>
    <Parameters>
        <Param id="ServiceID">ADSL01</Param>
        <Param id="ProductID">ADSL</Param>
    </Parameters>
</WSM::SetRequest>
```